

PRODUCTION- FRODUCTION-GRADE ARCHITECTURES

2026 TECHNICAL REFERENCE
REFERENCE

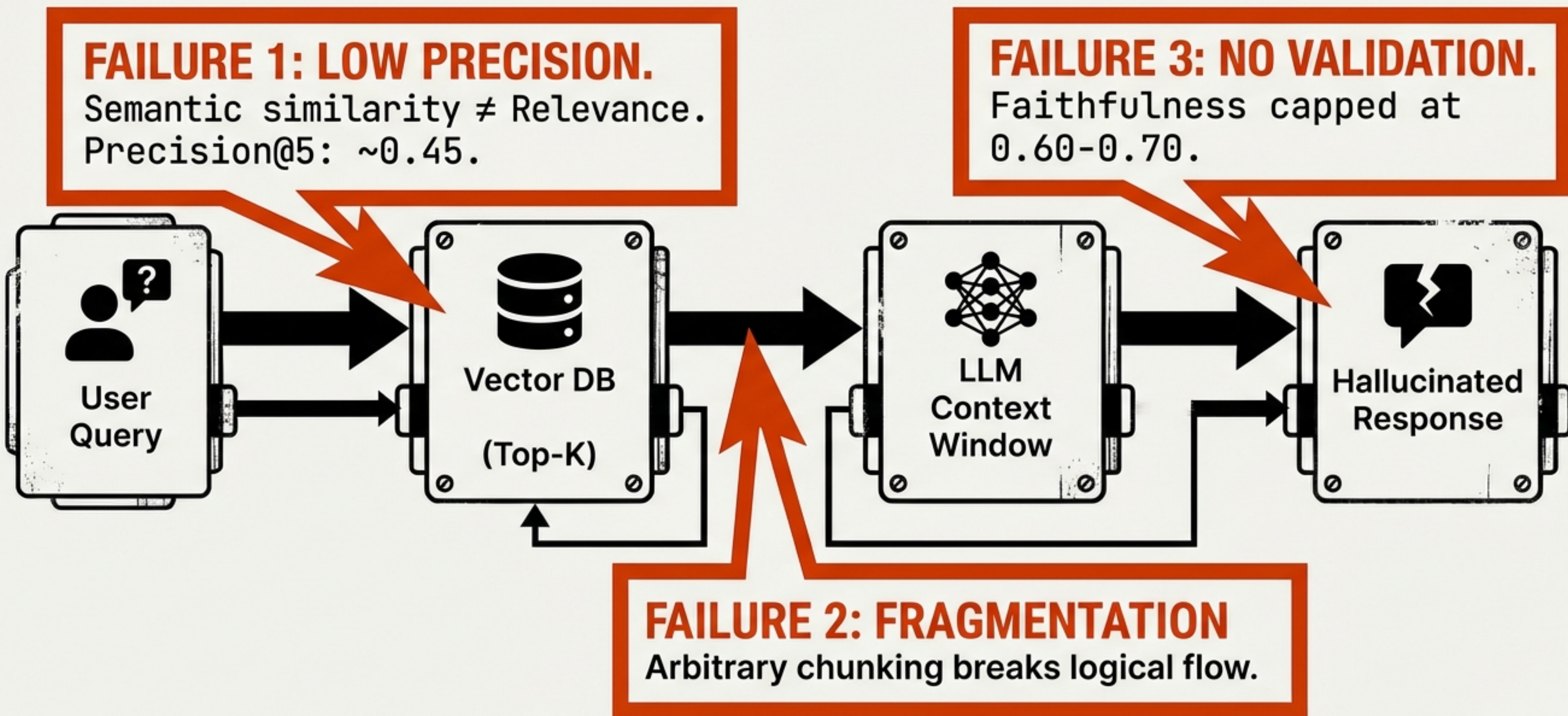
SUBTITLE: From Naive Retrieval
to Agentic Intelligence

TARGET AUDIENCE: AI
Architects, CTOs, Senior
Engineers

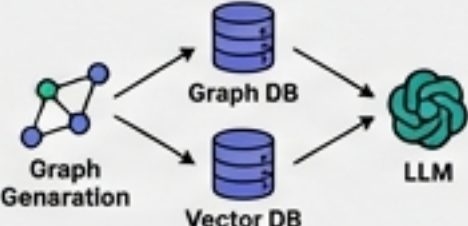
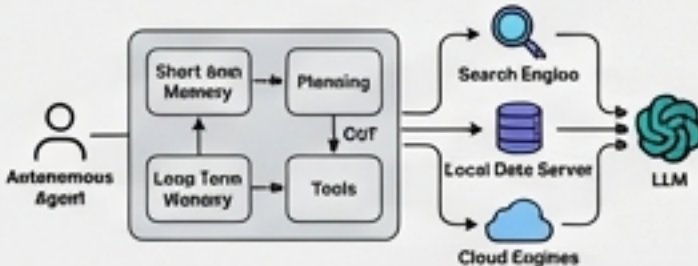
MARKET CONTEXT: \$40.34B
Projected by 2035 (35% CAGR)

CORE SHIFT: Static Retrieval
-> Reasoning Engines

THE REALITY CHECK: WHY NAIVE RAG FAILS



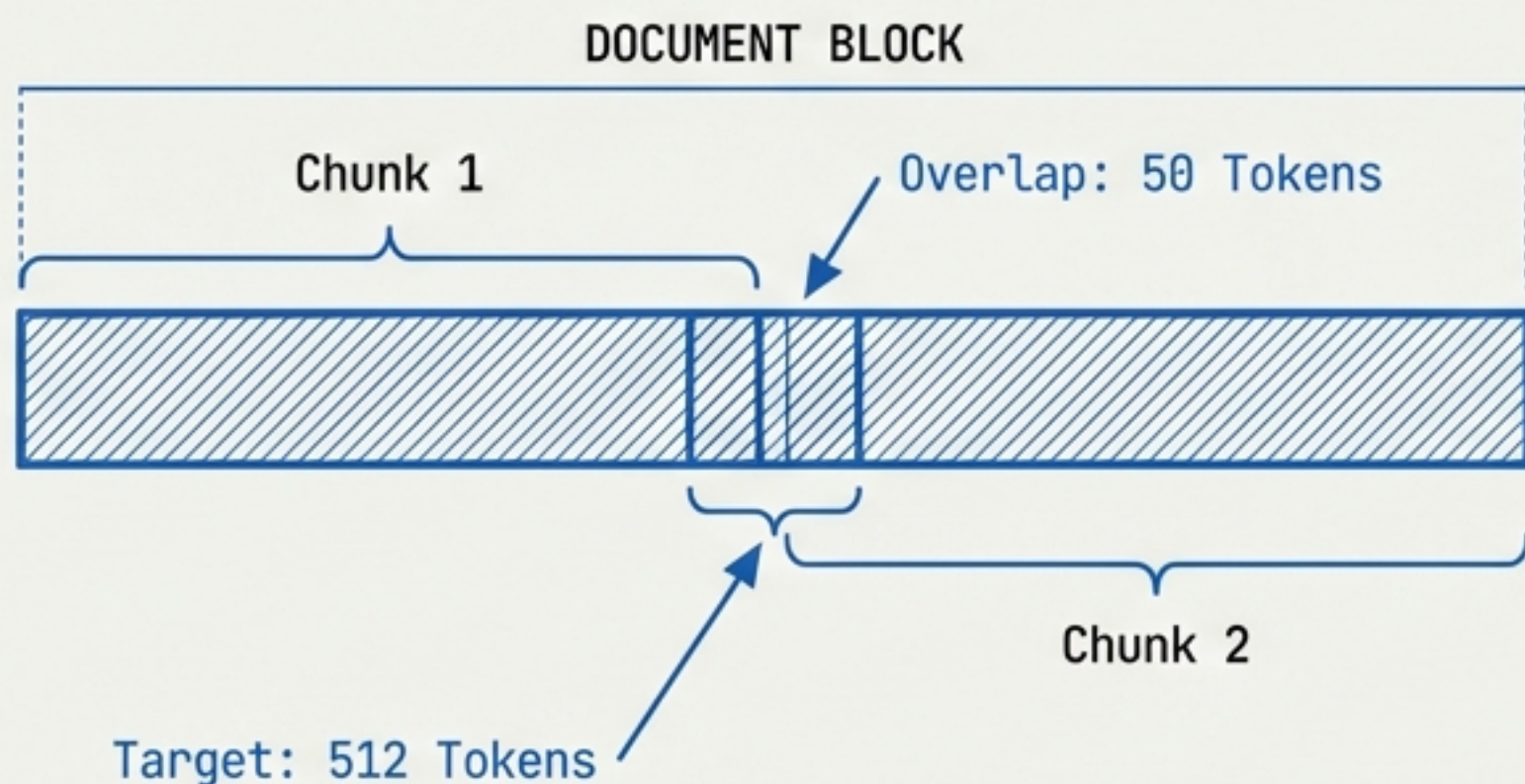
2026 ARCHITECTURE TAXONOMY

TIER	ARCHITECTURE	LATENCY	ACCURACY	USE CASE
Tier 0	Naive RAG (Vector Only) 	< 1s	0.60	Prototypes / MVP
Tier 1	Hybrid (Dense + Sparse) 	< 2s	0.80	Production Baseline
Tier 2	GraphRAG 	1.5 - 4s	0.85-0.92	Complex Relations
Tier 3	Corrective RAG (CRAG) 	+1 - 3s	~0.92	High Stakes (Finance)
Tier 4	Agentic RAG 	5 - 15s	0.95	Autonomous Research

TIER 1: THE INGESTION PIPELINE

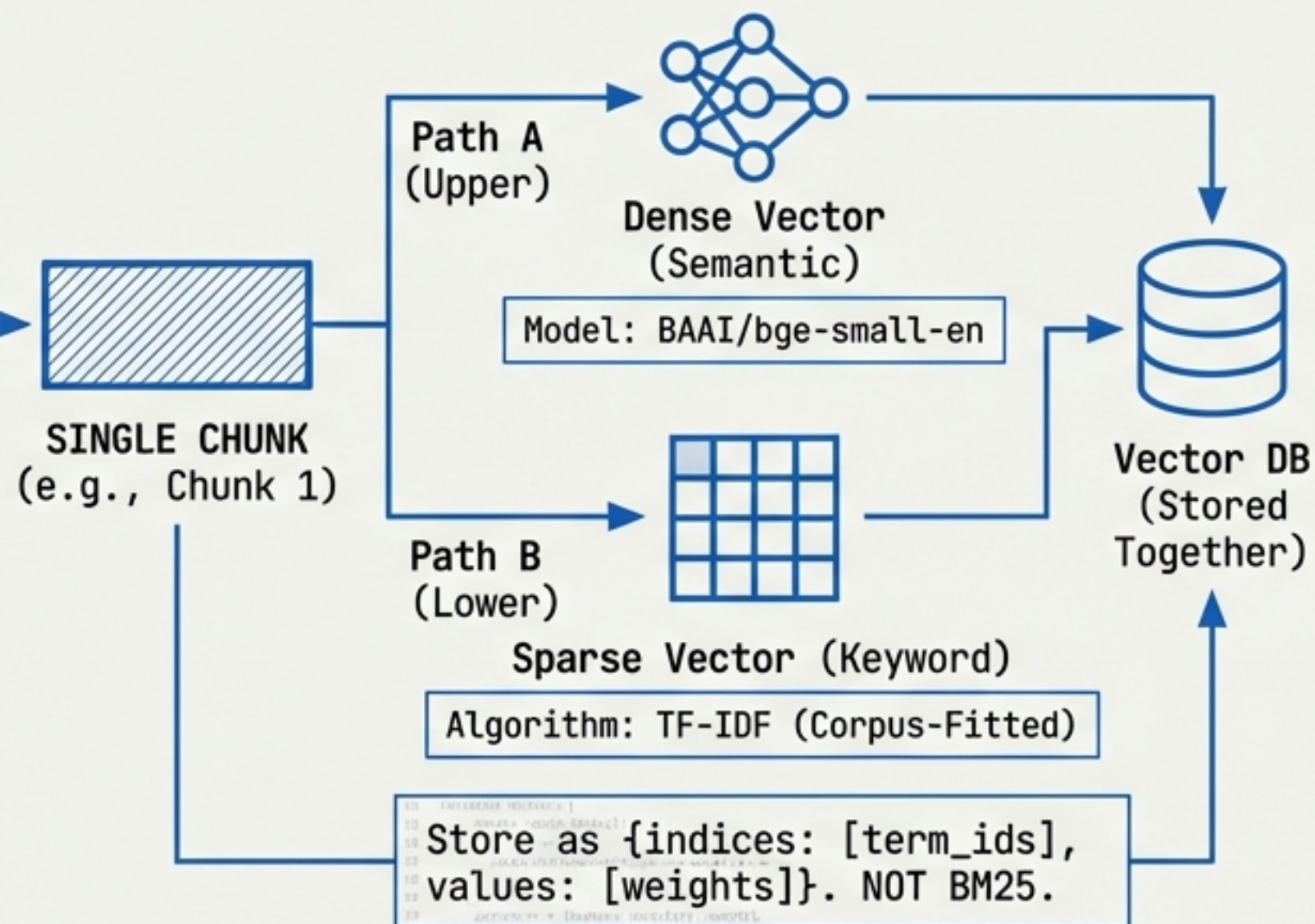
Solving Vocabulary Mismatch with Dual Embeddings

Token-Aware Chunking



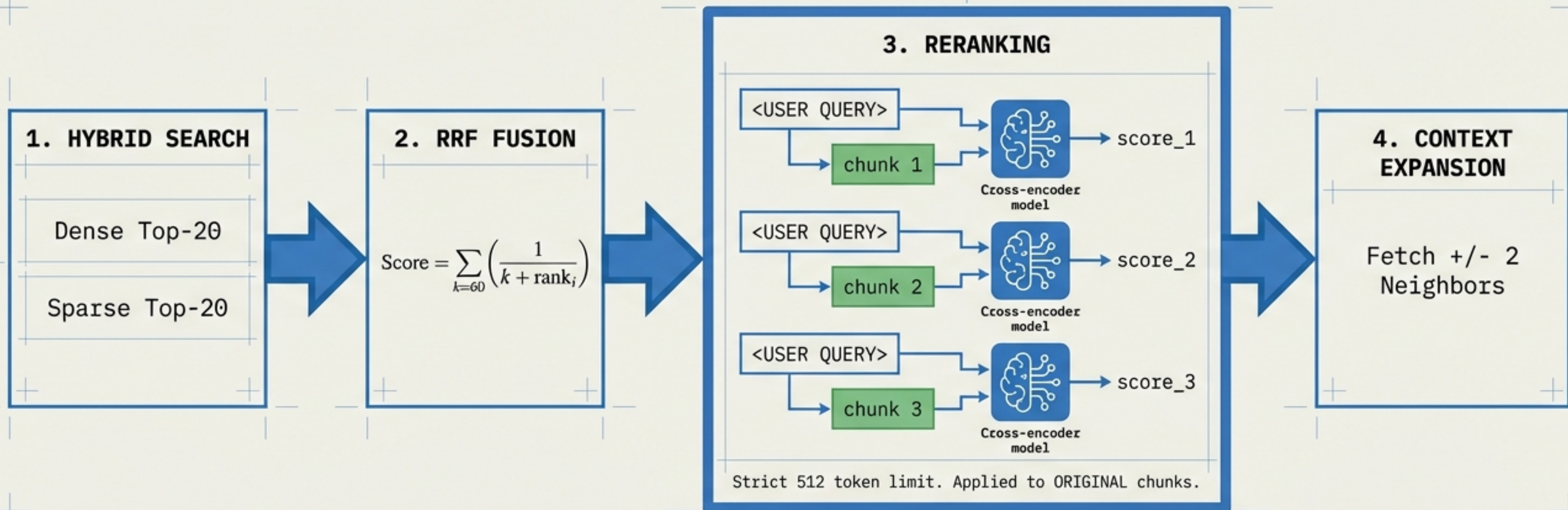
**Do NOT use character splitting.
Use tiktoken.**

Dual Embedding Path

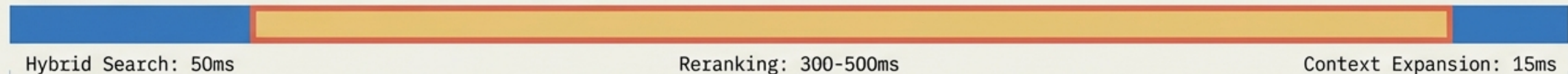


TIER 1: RETRIEVAL MECHANICS

The Golden Rule: Rerank Before You Expand

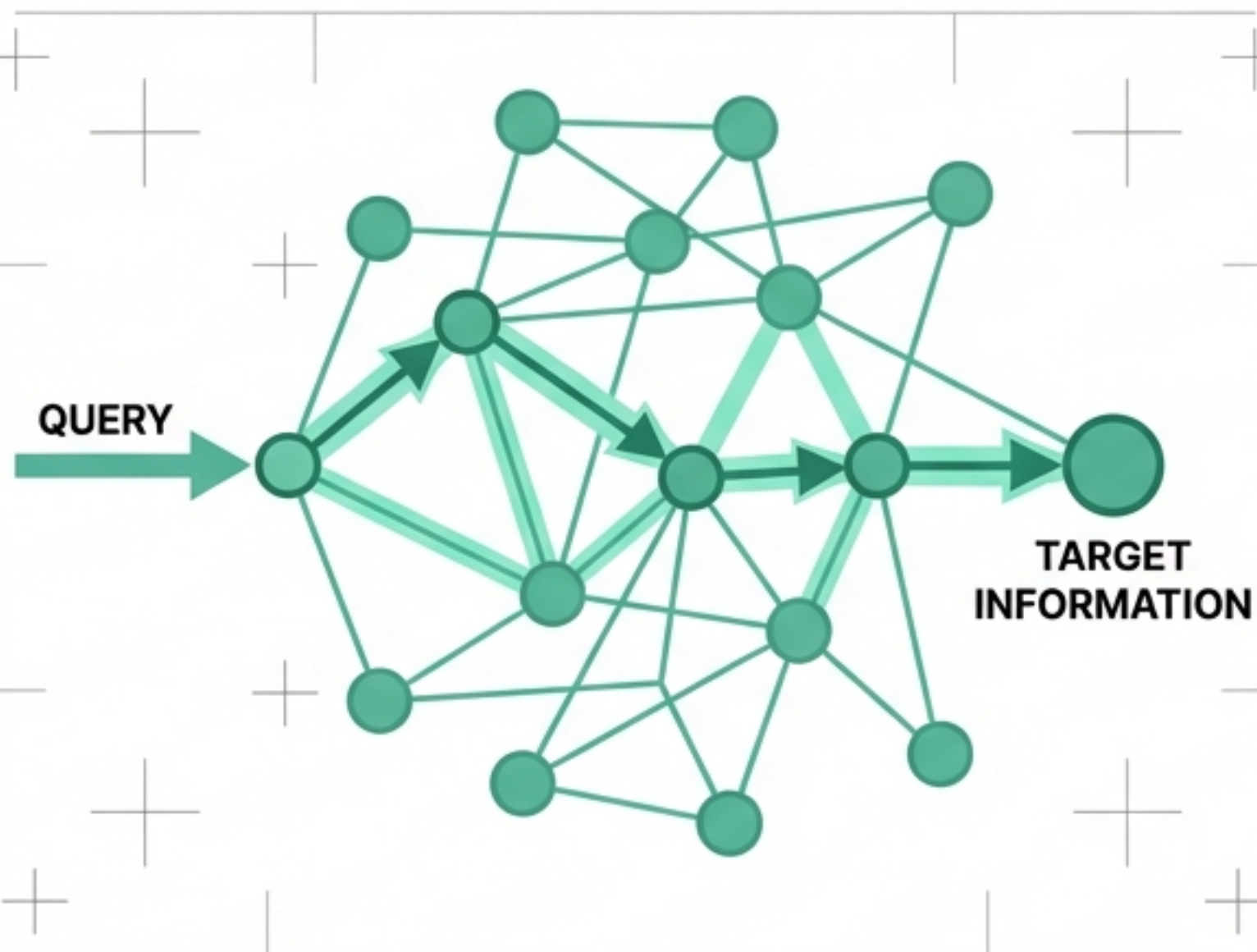
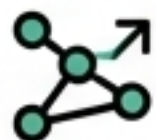


Latency Distribution (Total ~1.8s)



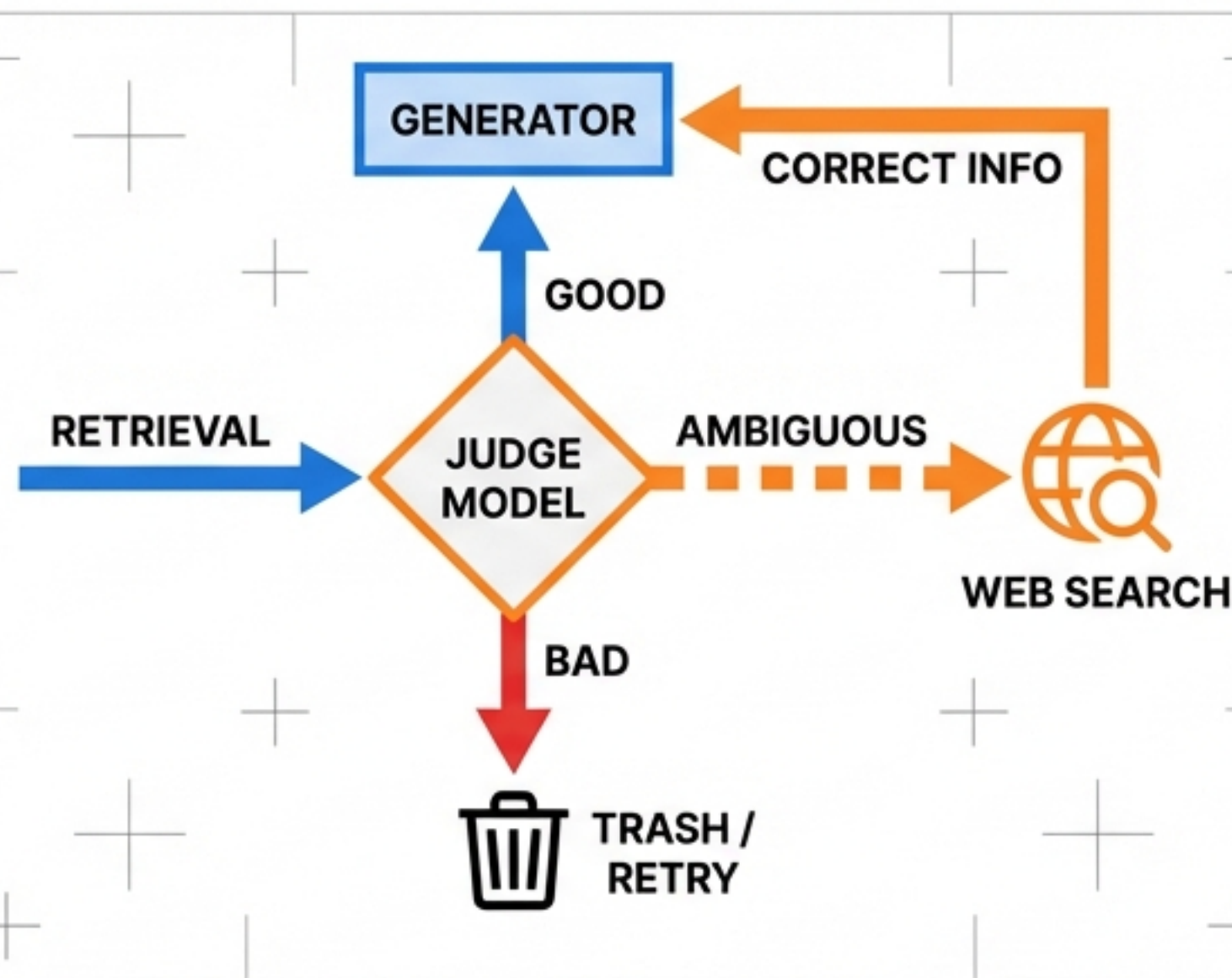
SPECIALIZED ARCHITECTURES (TIERS 2 & 3)

TIER 2: GraphRAG



Problem: Multi-hop reasoning. **Cost:** 3-5x Baseline.

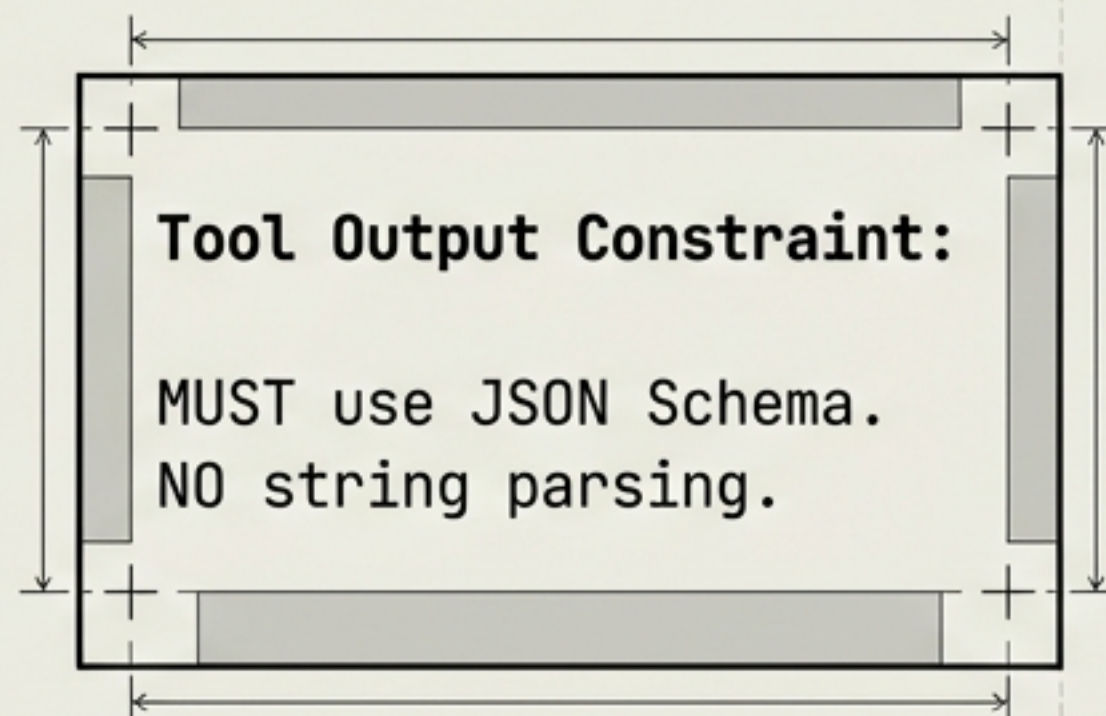
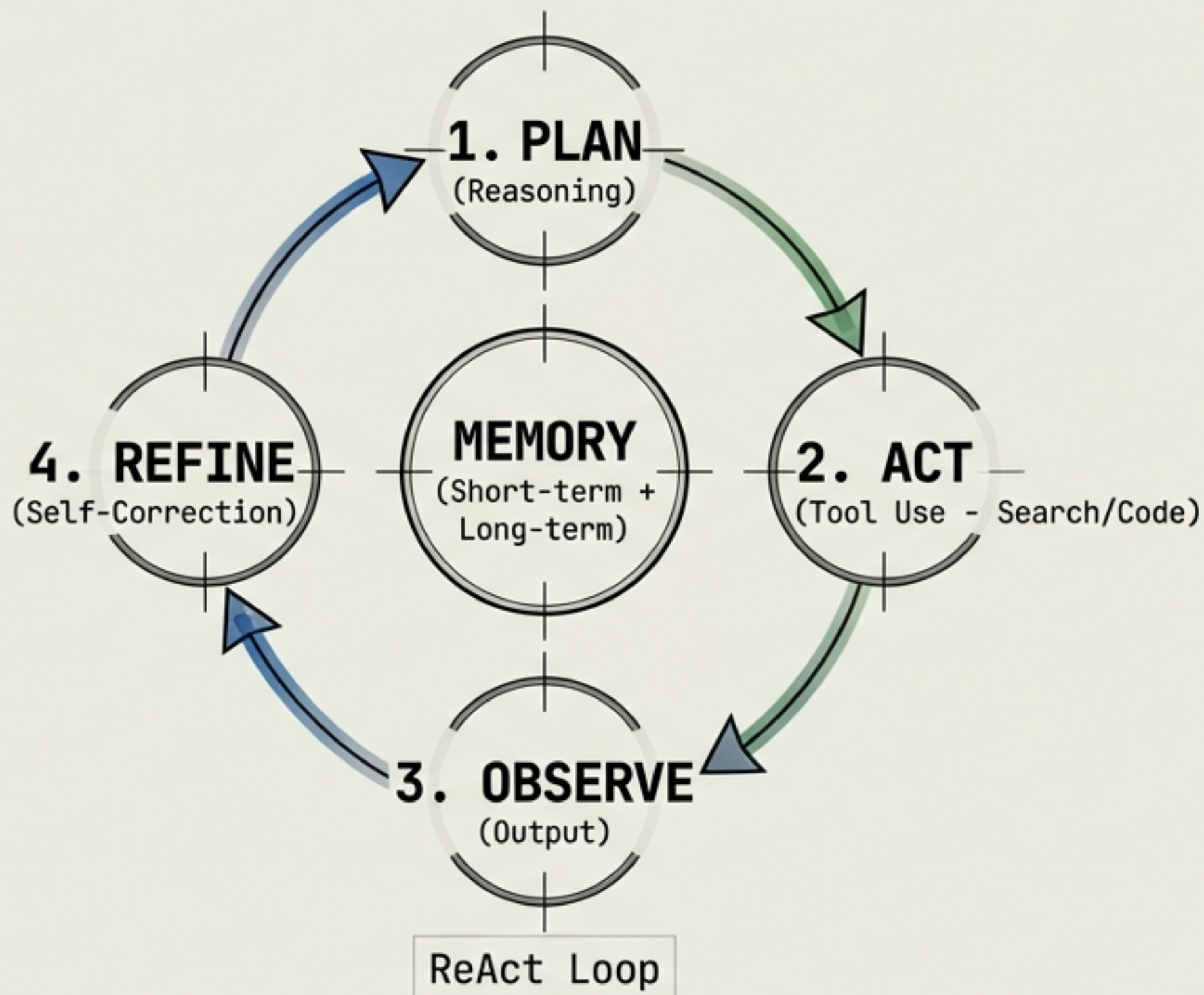
TIER 3: Corrective RAG (CRAG)



Problem: Blind trust. **Impact:** -60% Hallucinations.

TIER 4: AGENTIC ARCHITECTURE

The Second Brain: Plan, Execute, Verify



THE FINE-TUNING REALITY

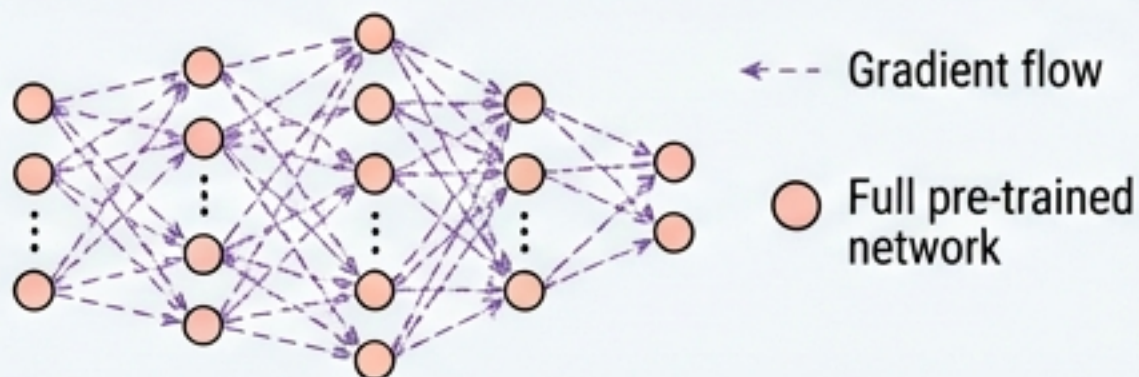


FINE-TUNING

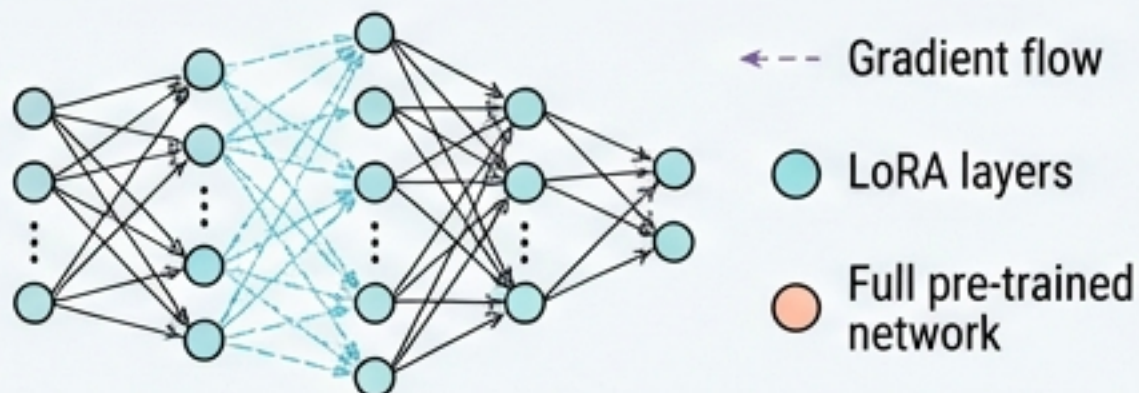
Updates **WEIGHTS**.

- **Solves:** Style, Tone, Format, Terminology.
- **Does NOT fix:** Facts/Grounding.

FULL FINE TUNING



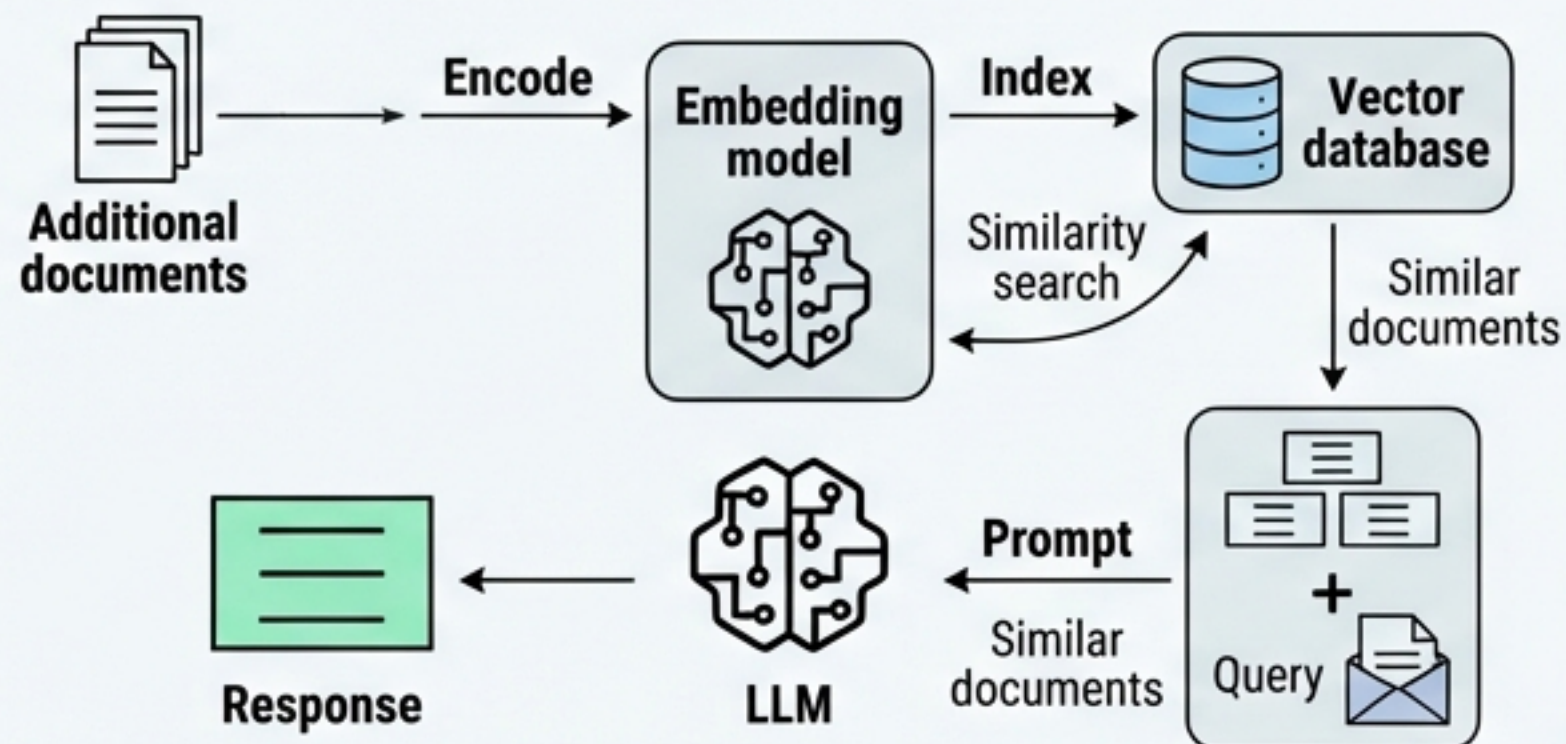
LoRA FINE TUNING



RAG

Updates **CONTEXT**.

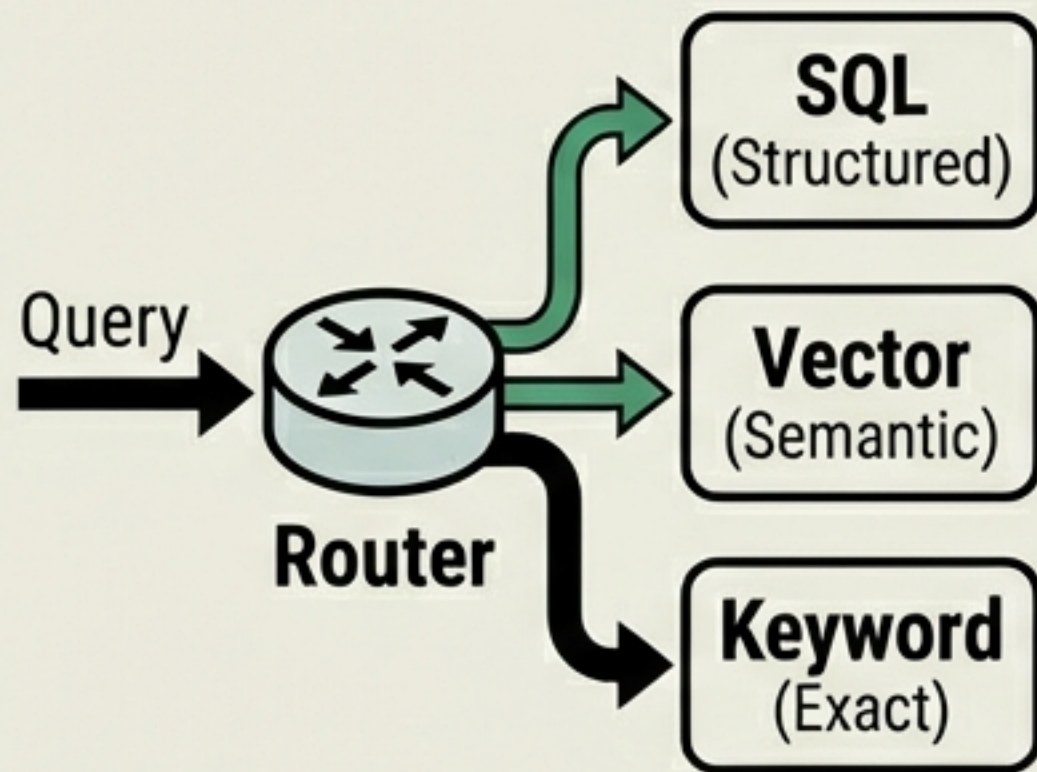
- **Solves:** Facts, Current Data, Specificity.



Recommendation: Use LoRA (Low-Rank Adaptation) for style, RAG for truth.

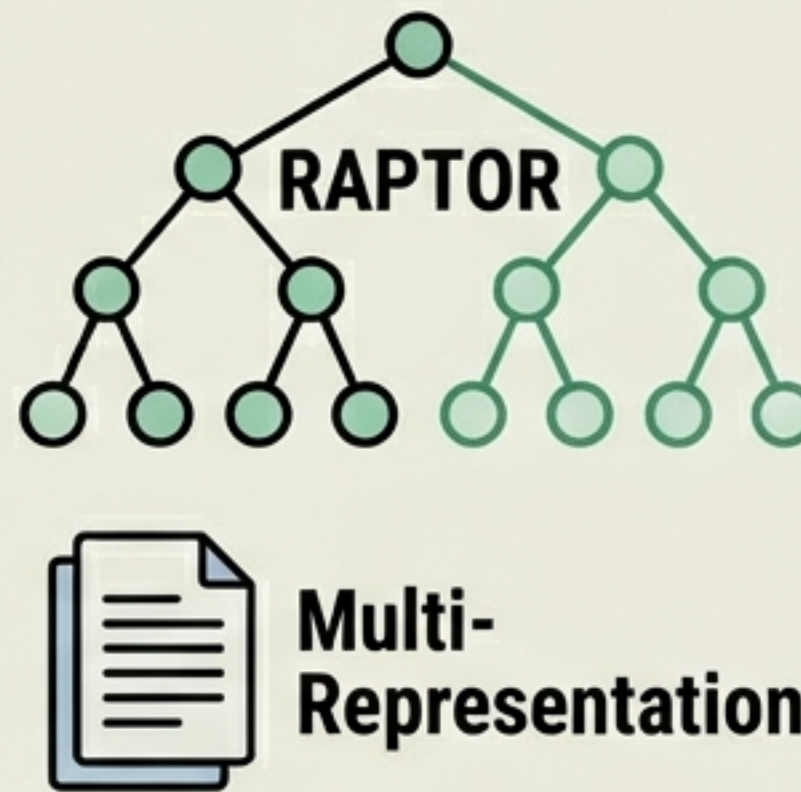
HIGH-ROI ENTERPRISE OPTIMIZATIONS

INTELLIGENT ROUTING



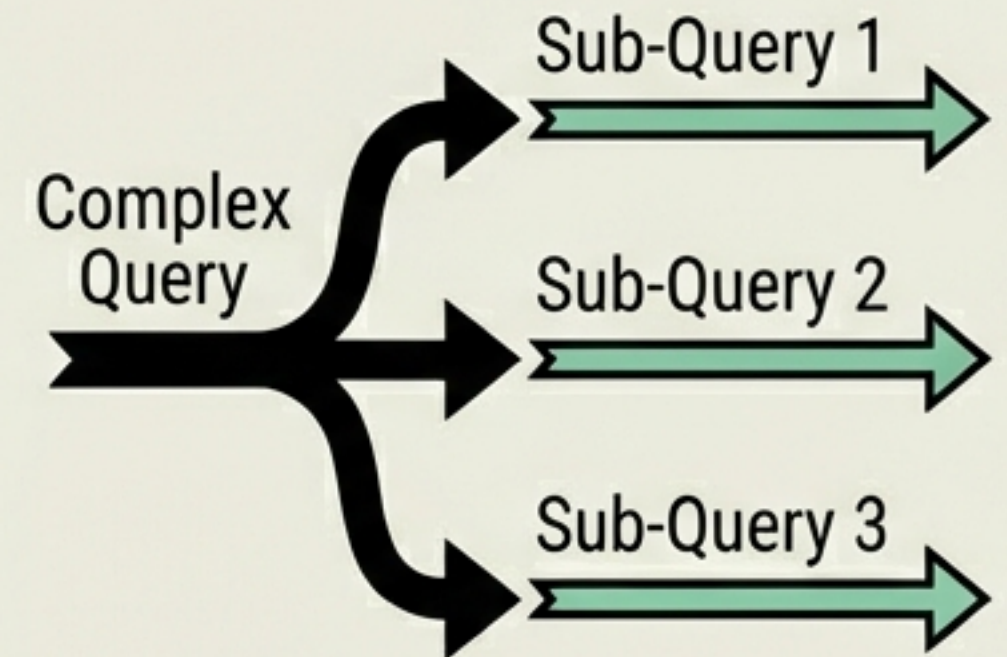
**Cuts cost/latency
by 40%.**

ADVANCED INDEXING



Cluster thematic queries
& index summaries.

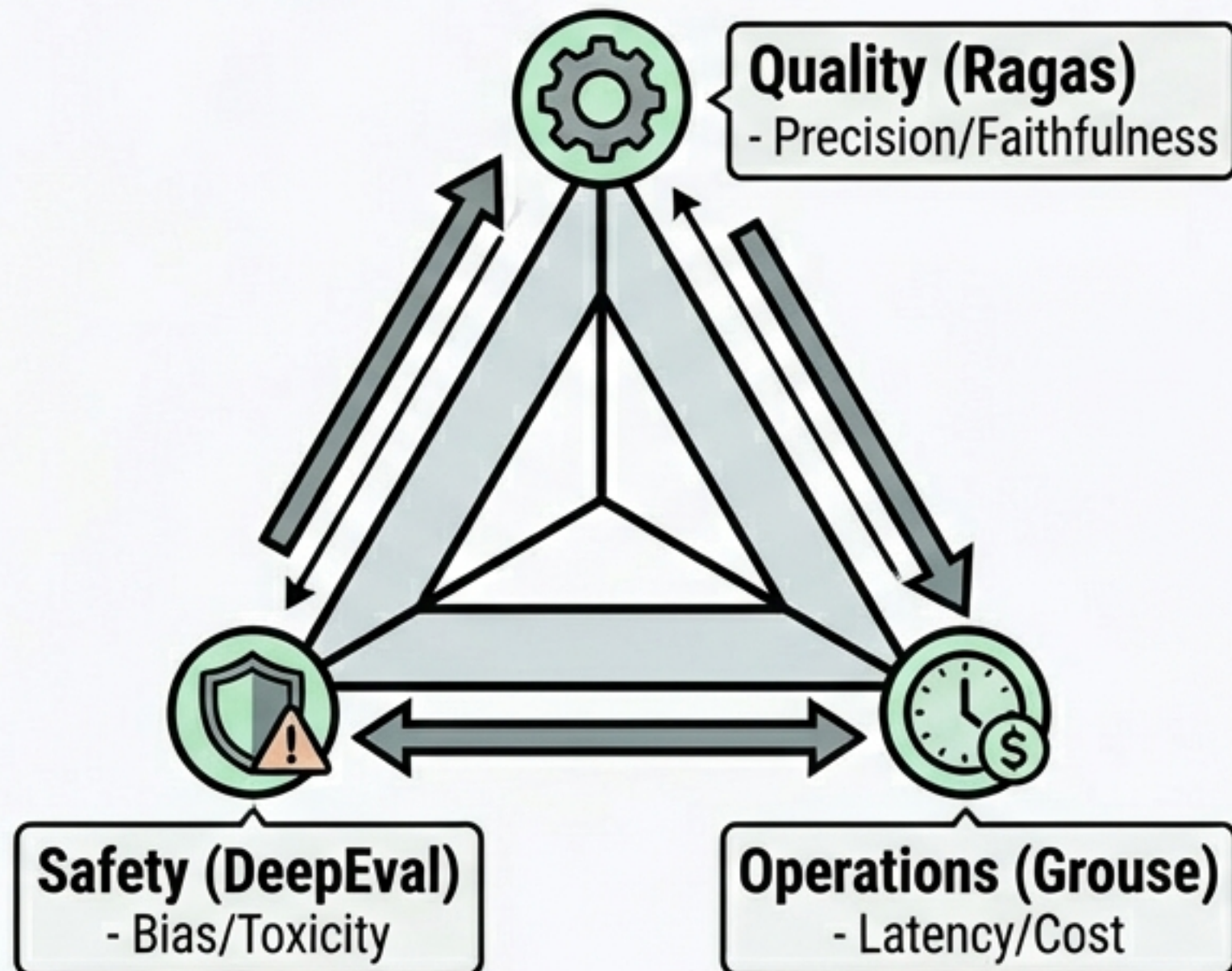
QUERY TRANSFORMATION



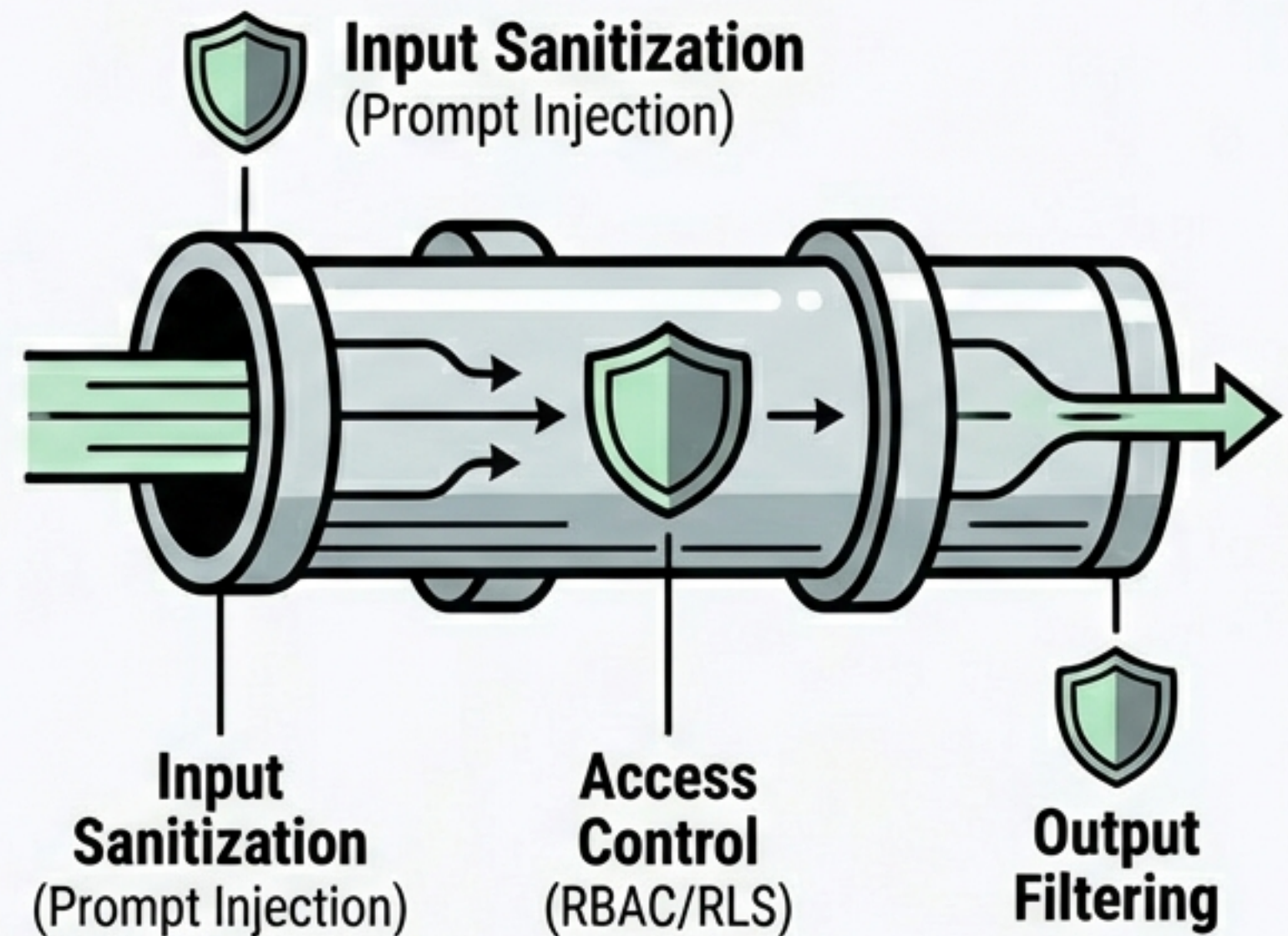
Decomposition &
Step-back prompting.

EVALUATION & SECURITY FRAMEWORKS

The Evaluation Triad



Security Defense Layer

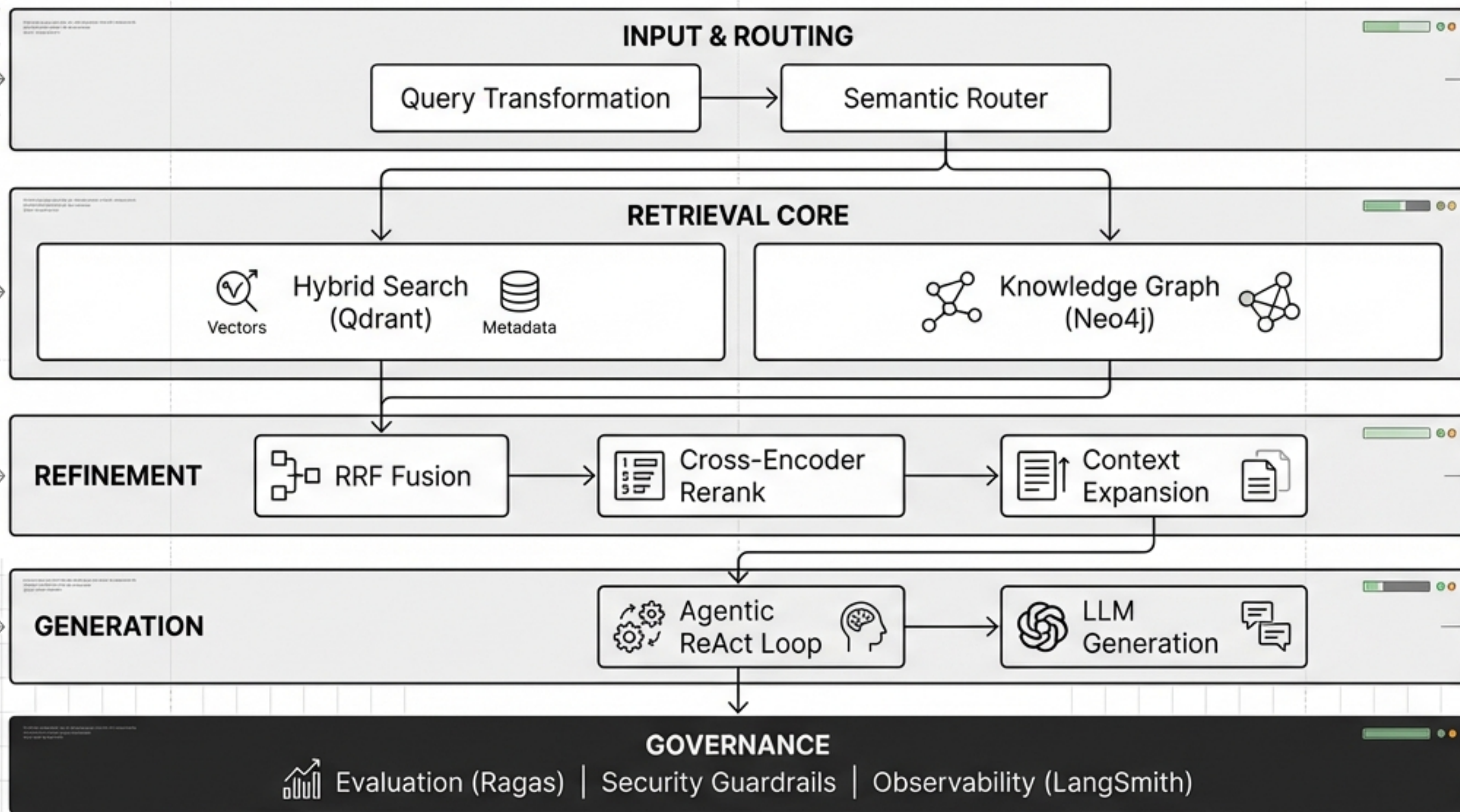


TECHNICAL ANTI-PATTERNS

Corrections for Production Systems

✗	BM25 for Indexing	✓	BM25 is Query-Time Only. Use TF-IDF/SPLADE.
✗	Expand Before Rerank	✓	Rerank Original Chunks (512 limit), THEN Expand.
✗	Character Chunking	✓	Token-Aware Chunking (tiktoken).
✗	String-Based Parsing	✓	JSON Mode / Function Calling.
✗	Fine-Tuning for Facts	✓	Fix Retrieval Pipeline.

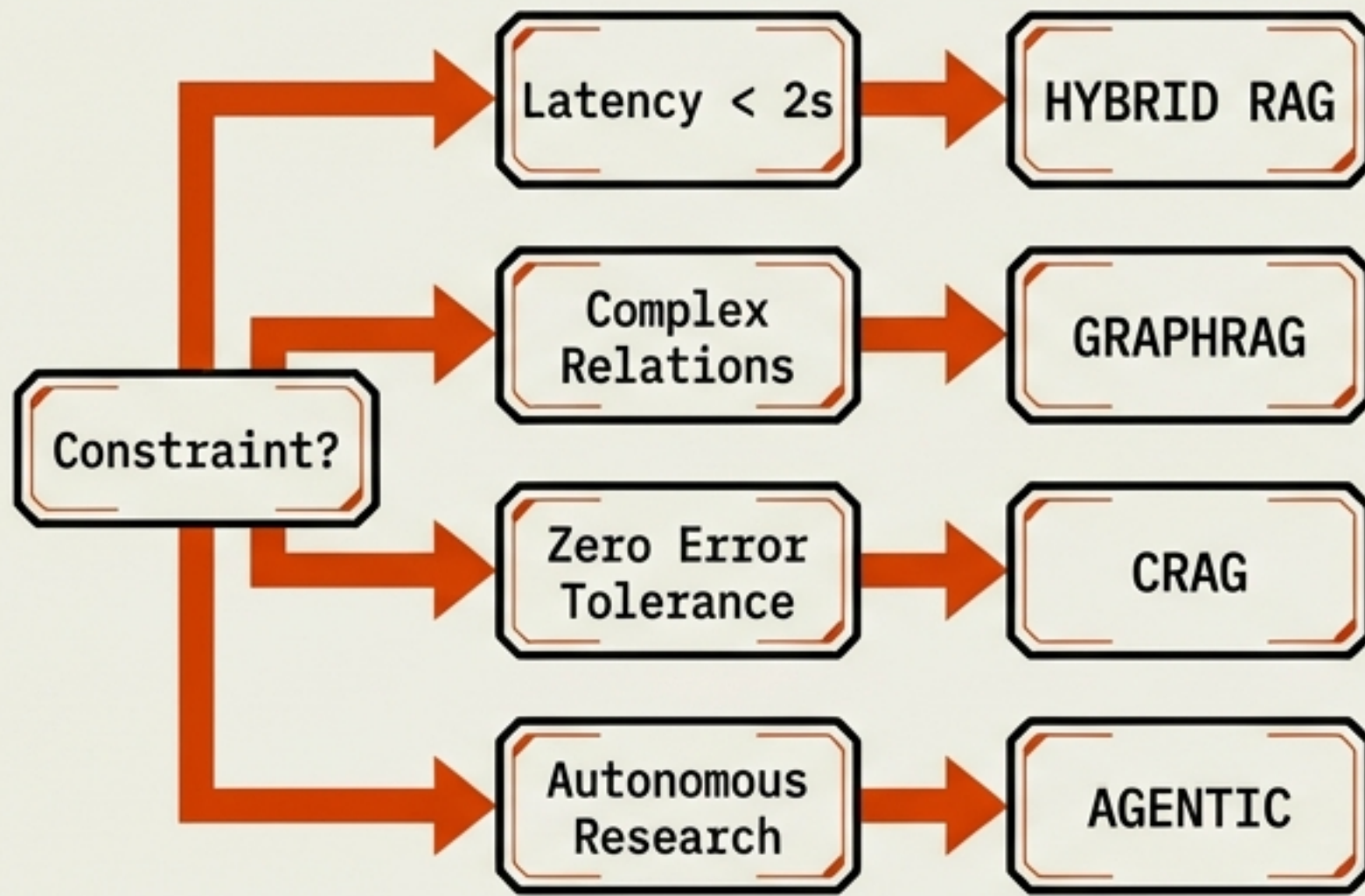
REFERENCE ARCHITECTURE 2026



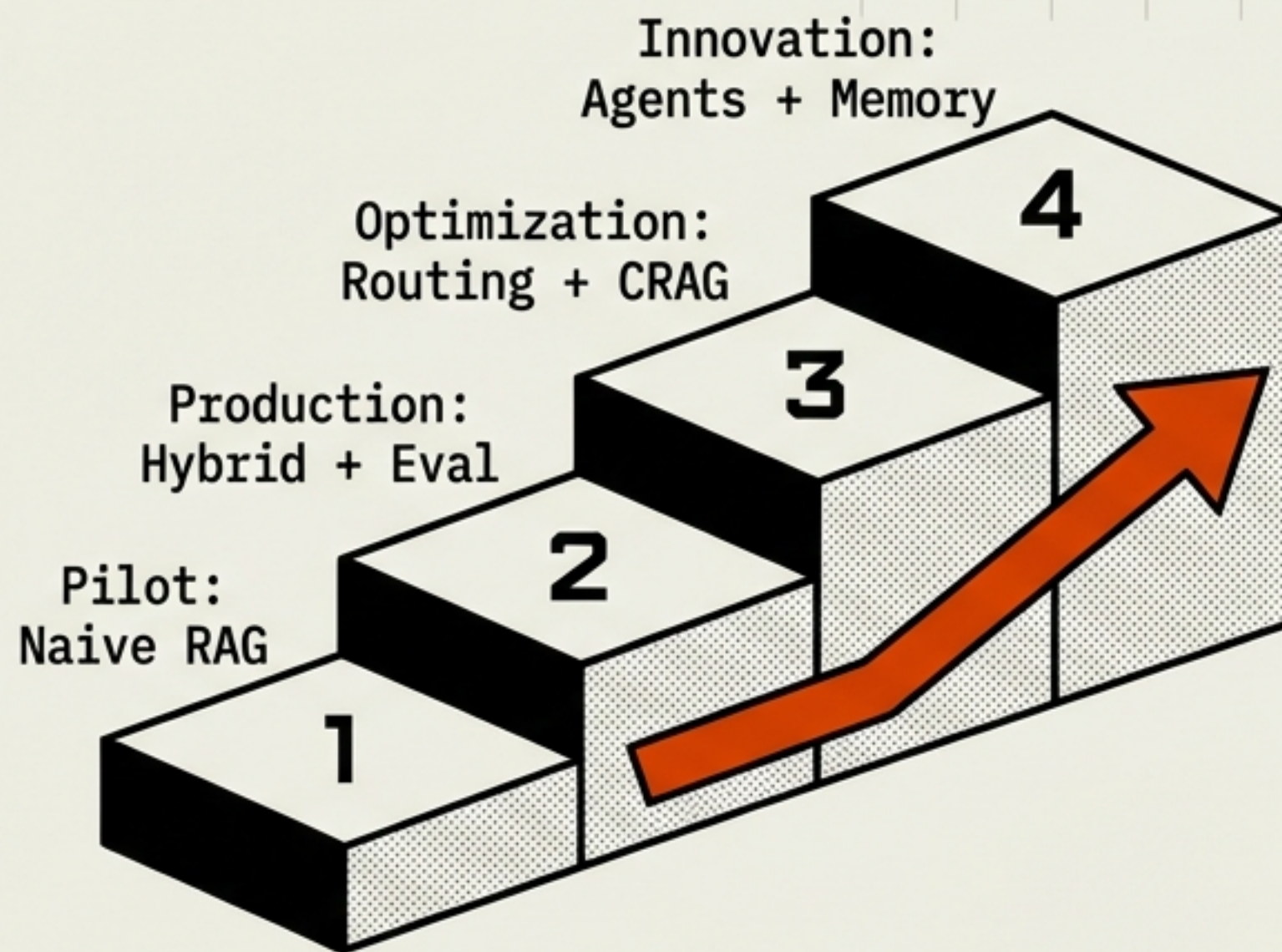
DECISION FRAMEWORK & MIGRATION

Strategic Choices & Phased Implementation

Architectural Decision Matrix



4-Phase Migration Path



PRODUCTION HARDENING CHECKLIST

PRODUCTION READINESS CHECK

1. PERFORMANCE

- ☒ Rate Limiting (10 q/min)
- ☒ Semantic Caching (Redis) - 40% Latency Drop

2. RELIABILITY

- ☒ Keyword Fallback (if Vector fails)
- ☒ Async Ingestion Workers

3. OBSERVABILITY

- ☒ Trace Every Step (LangSmith)
- ☒ Monitor 'Expensive Queries'

Don't build a custom Vector DB. Don't train your own base model.

THE FUTURE IS INFRASTRUCTURE

In 2026, the competitive advantage is not the model.
It is the **ARCHITECTURE**.



“Build for resilience, not just intelligence.”